

# AI4Math: Scientific Computing and Mathematical AI Portfolio

Python, PyTorch, C/C++, LaTeX, Jupyter, PyVista/VTK | 2026

**Project positioning.** A course-based computational mathematics portfolio demonstrating practical preparation for research assistant work: deep-learning experiments, numerical simulation, scientific visualization, reproducible reports, and transparent AI-assisted development.

## Selected technical evidence

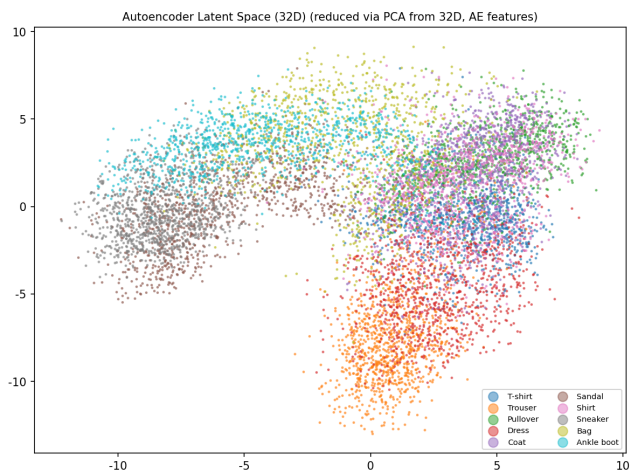
- Implemented PyTorch image-learning pipelines on MNIST-family datasets, including MLP/CNN baselines and AE/DAE/VAE latent-space visualizations.
- Built numerical computing modules in C/C++ and Python, including sparse CRS operations, LU/Poisson solver demos, and convergence checks.
- Created scientific visualization demos using Jupyter, PyVista, and VTK, including MRI slicing, terrain rendering, molecule density, and weather fields.
- Completed a celestial-mechanics final project with N-body integration, cached JPL Horizons validation, lunar flyby modeling, launch-window search, sensitivity analysis, and a reproducible LaTeX report.

## Representative deliverables

- `final_project/Qian/report.pdf`
- `final_project/Qian/README.md`
- `week11/experiments.md`
- `week13/outputs/opendx_cases/case4_mri_slices.png`

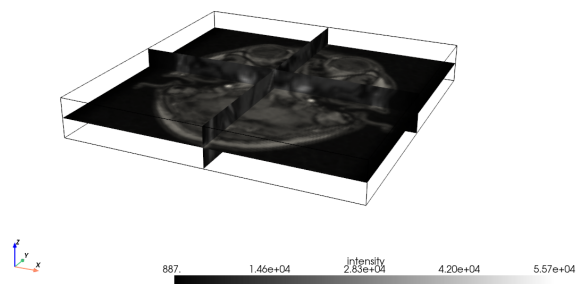
## RA-relevant skills

|                              |                                                                                   |
|------------------------------|-----------------------------------------------------------------------------------|
| <b>Programming</b>           | Python, C/C++, Git, Makefile workflows                                            |
| <b>ML/AI</b>                 | PyTorch, CNNs, autoencoders, experiment logging                                   |
| <b>Numerics</b>              | N-body dynamics, RK4, Verlet, sparse matrices                                     |
| <b>Visualization Writing</b> | matplotlib, PyVista, VTK, Jupyter LaTeX reports, result tables, AI-use disclosure |

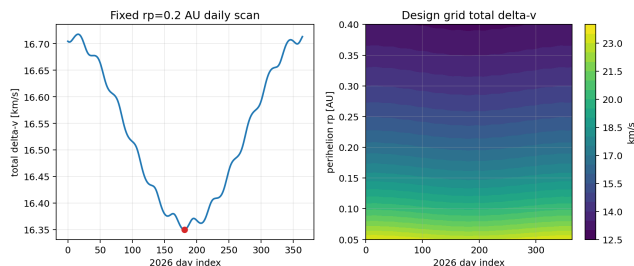


Autoencoder latent-space visualization

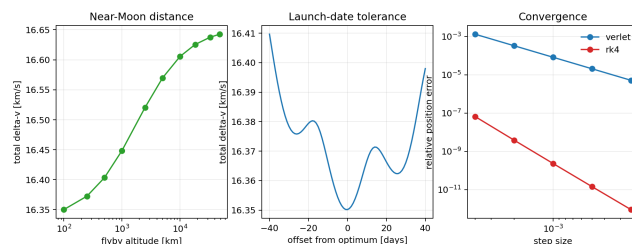
MRI orthogonal slices



MRI slice visualization with PyVista/VTK



2026 launch-window scan



Sensitivity and convergence analysis

**Public-release note.** Local credentials, private API tokens, temporary build files, executable binaries, and cache directories are excluded or replaced with placeholders.